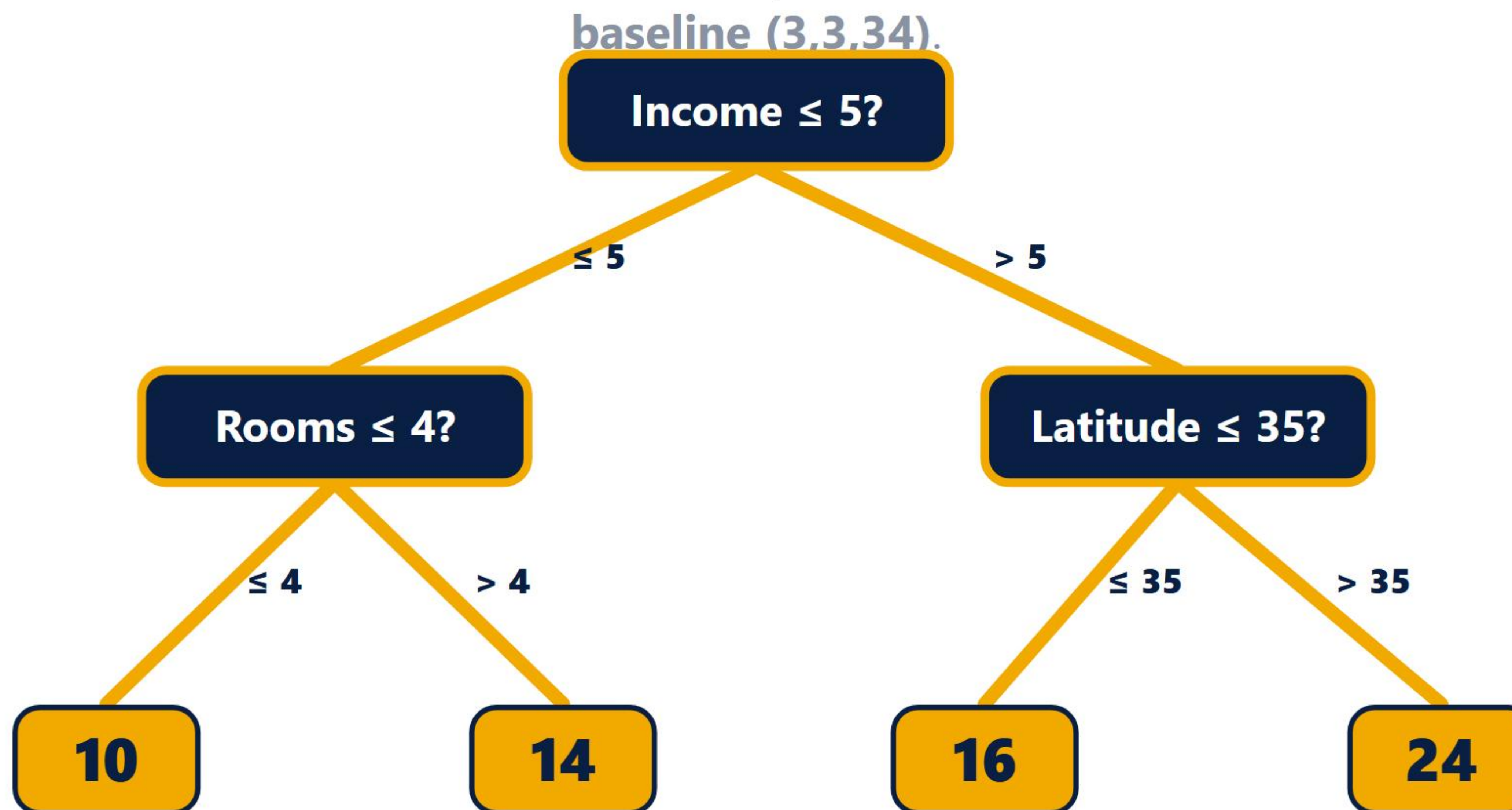


The model: one shallow tree. We trace all 8 coalitions — present features use the **home (8,6,38)**, absent use the **baseline (3,3,34)**.

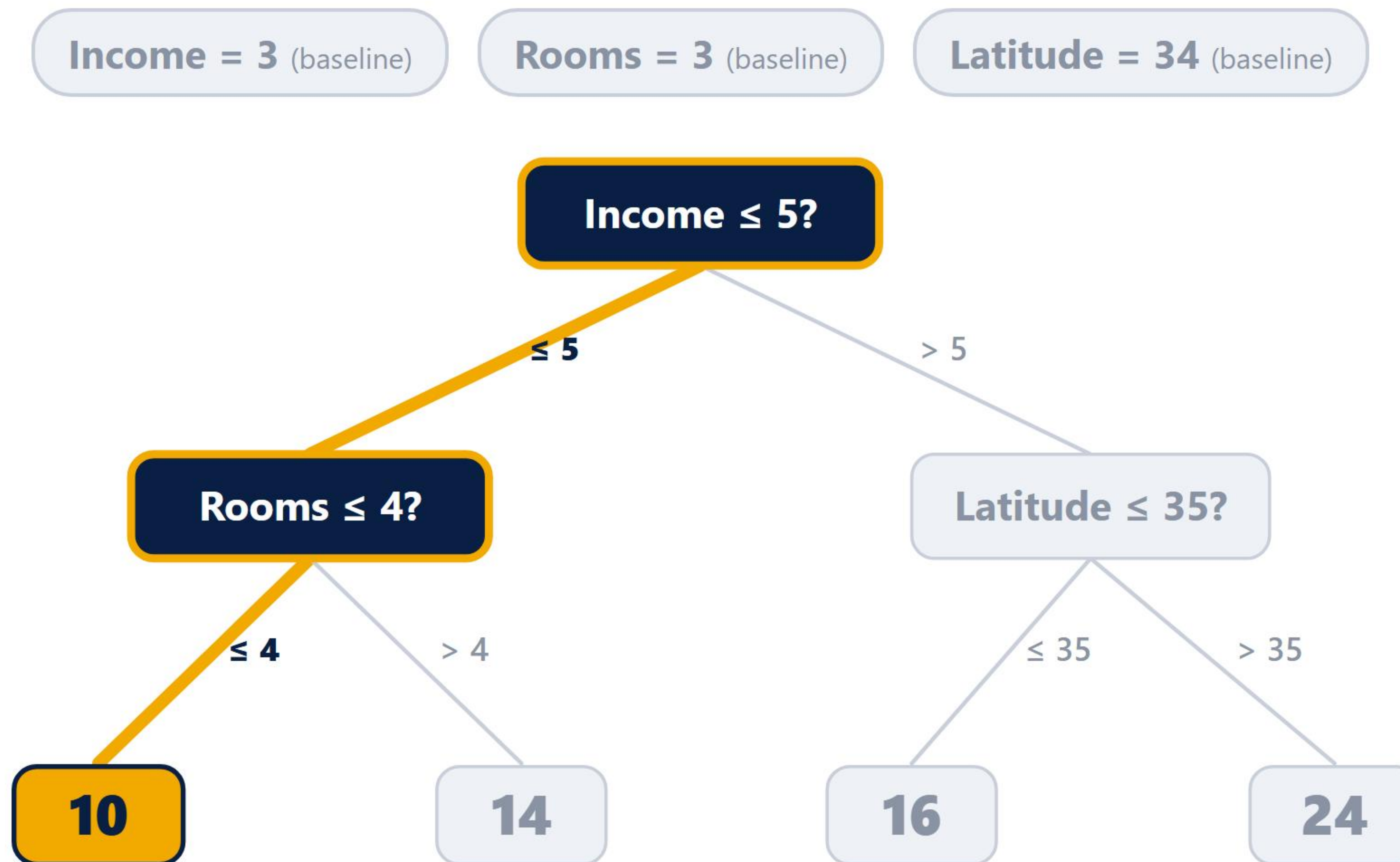


Each coming slide lights the path for one coalition and reads off $v(S)$.

COALITION

$S = \{ \}$ (baseline)

$v(S) = 10$



Nobody known — every feature on the baseline row. This is where every explanation starts.

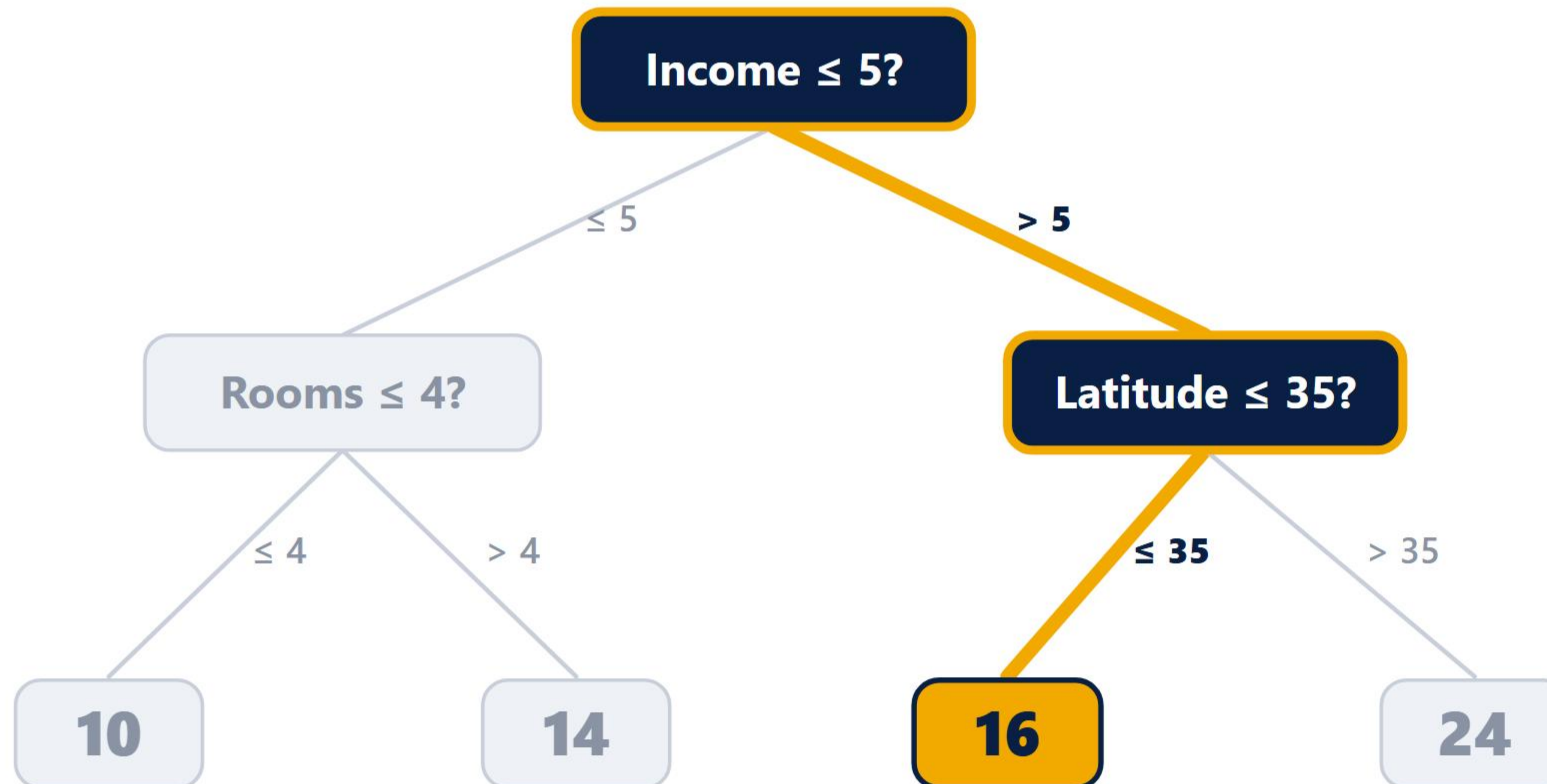
COALITION
S = {I}

v(S) = 16

Income = 8 (home)

Rooms = 3 (baseline)

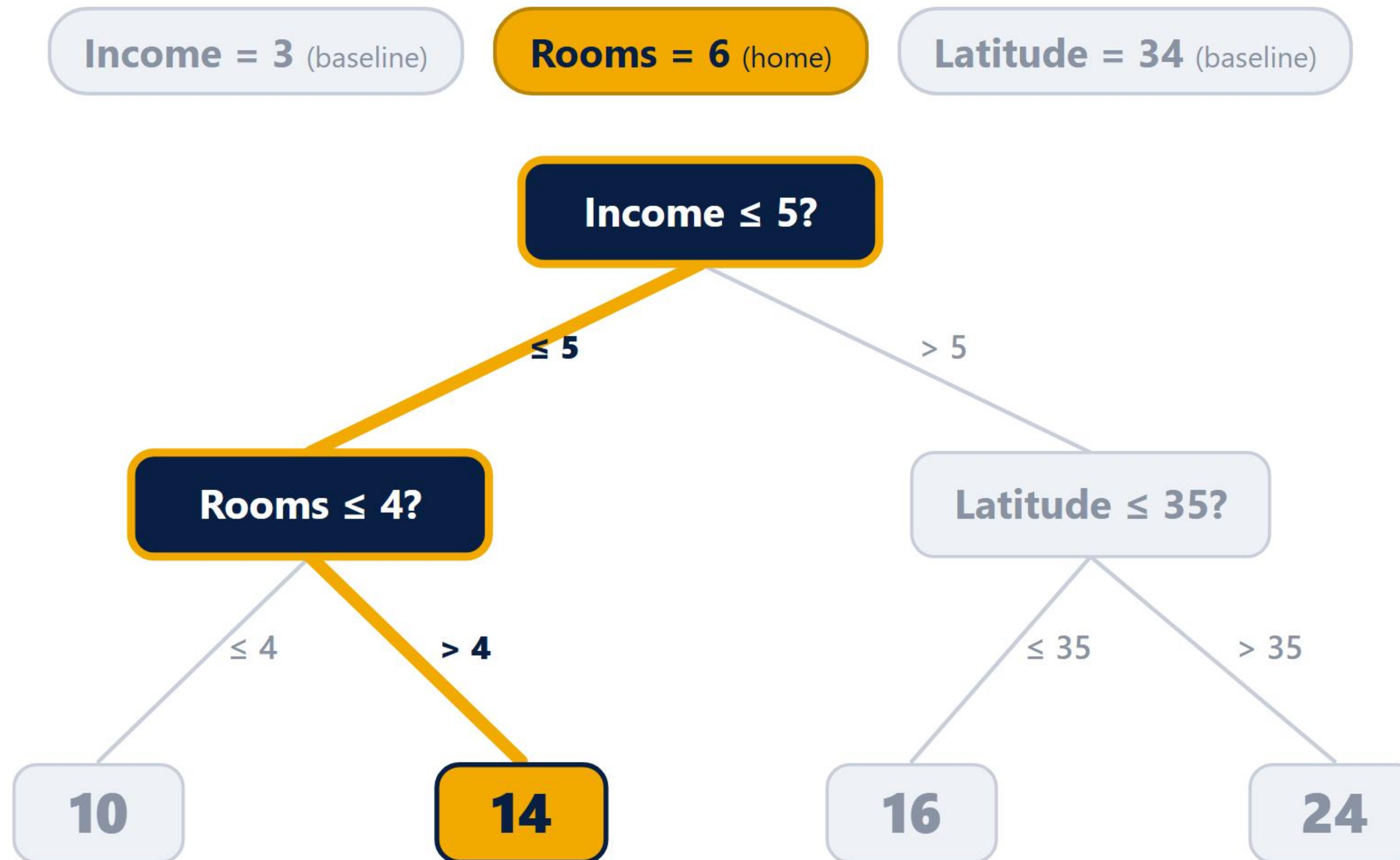
Latitude = 34 (baseline)



Income alone flips us onto the high-income branch.

COALITION
S = {R}

v(S) = 14



Rooms known, but income still baseline (low) $\rightarrow$ the rooms split decides.

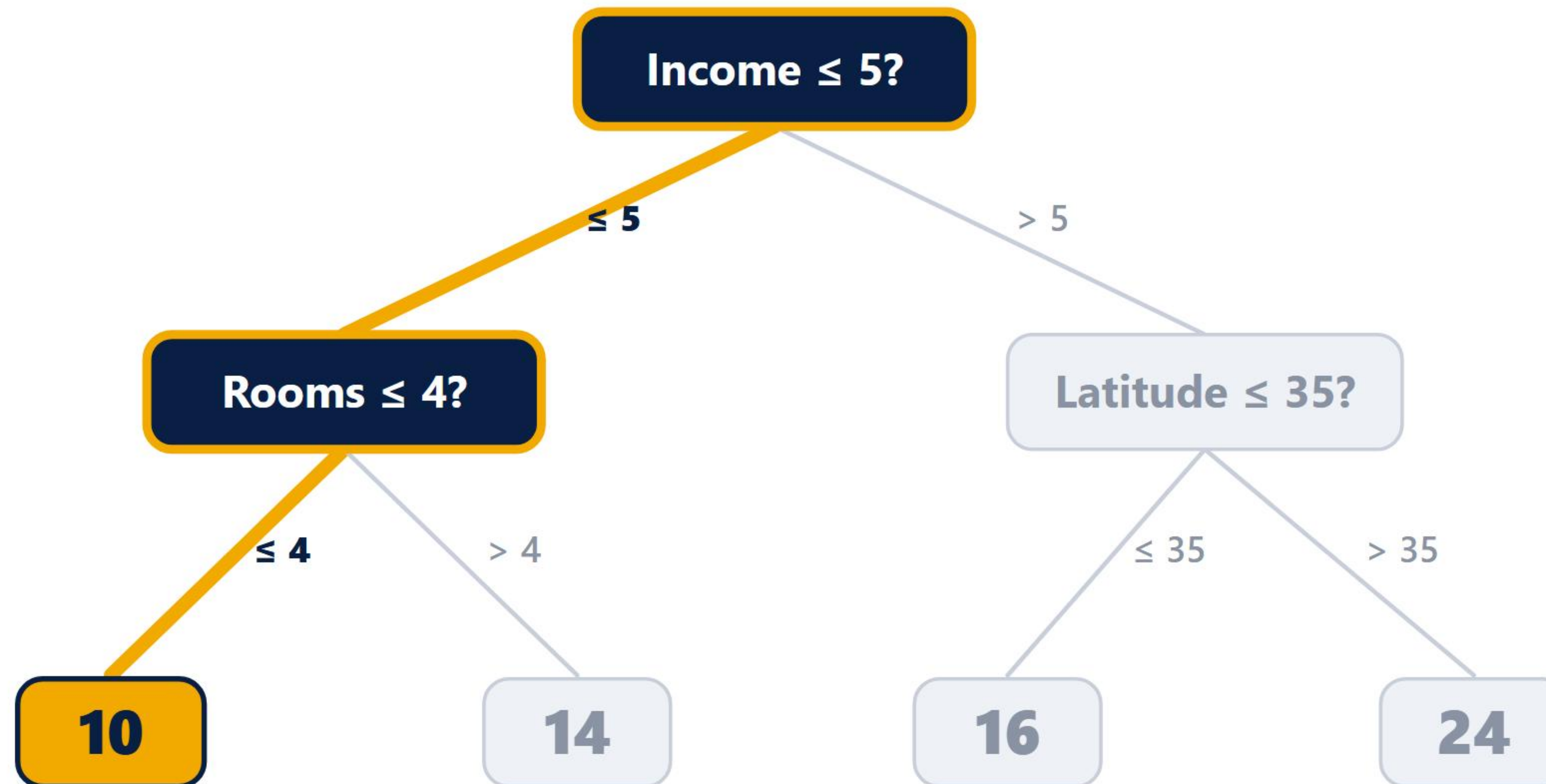
COALITION
S = {L}

v(S) = 10

Income = 3 (baseline)

Rooms = 3 (baseline)

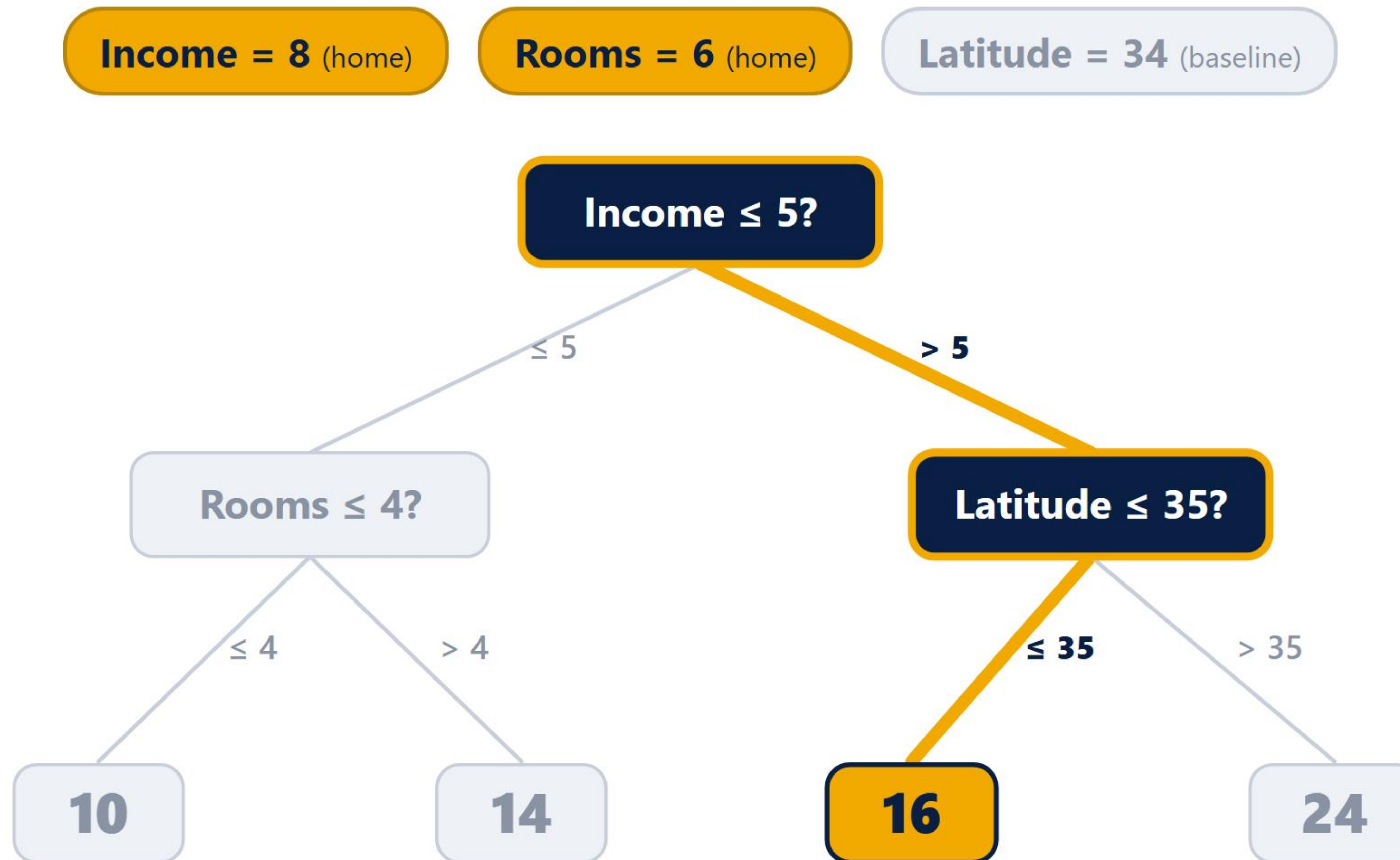
Latitude = 38 (home)



Latitude is never checked on the low-income branch → same leaf as baseline → latitude adds nothing HERE.

COALITION
S = {I, R}

v(S) = 16



Rooms is never checked on the high-income branch → knowing rooms adds nothing once income is high (an interaction).

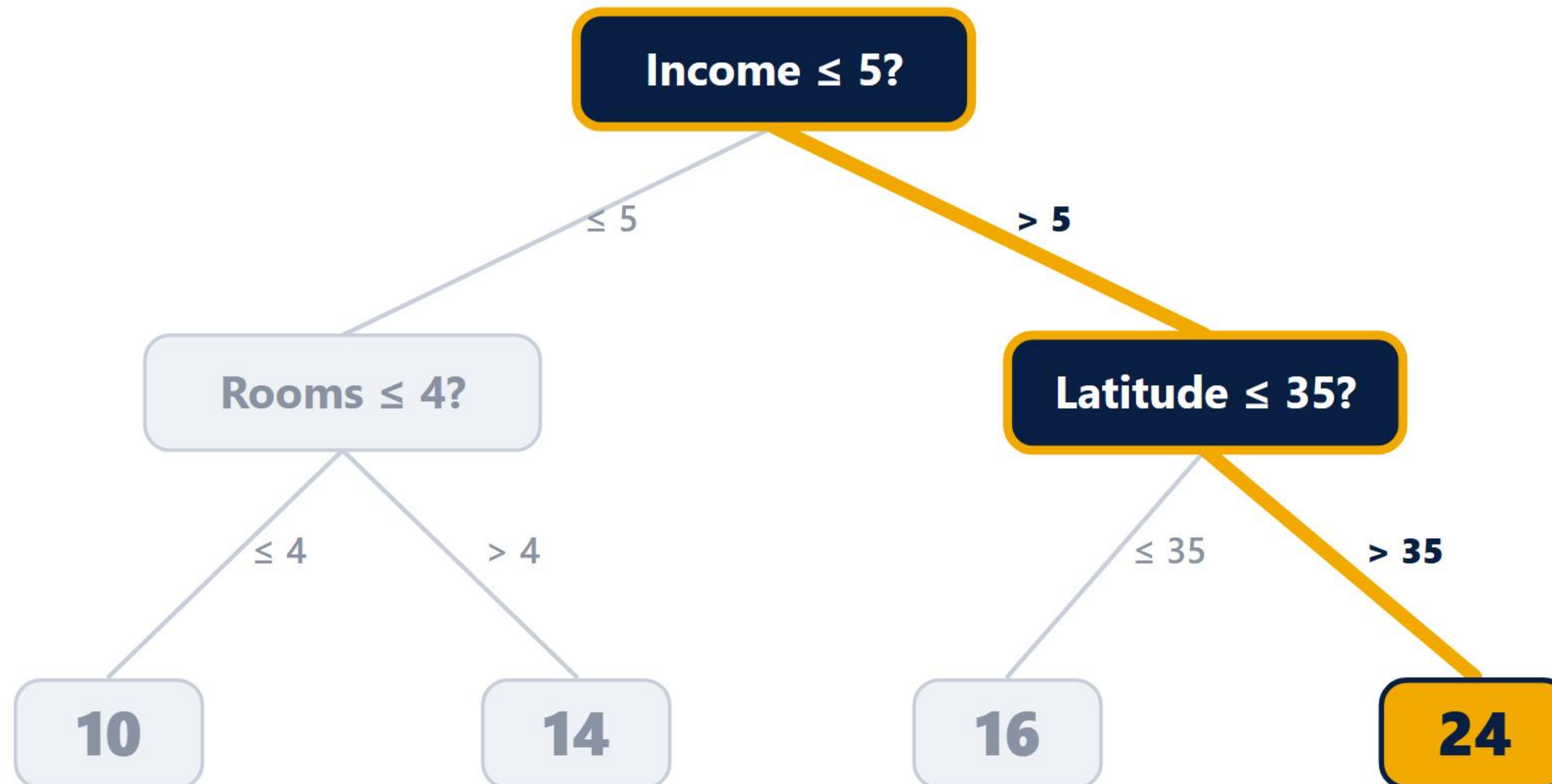
COALITION
S = {I, L}

v(S) = 24

Income = 8 (home)

Rooms = 3 (baseline)

Latitude = 38 (home)



Income high AND latitude high → the top leaf.

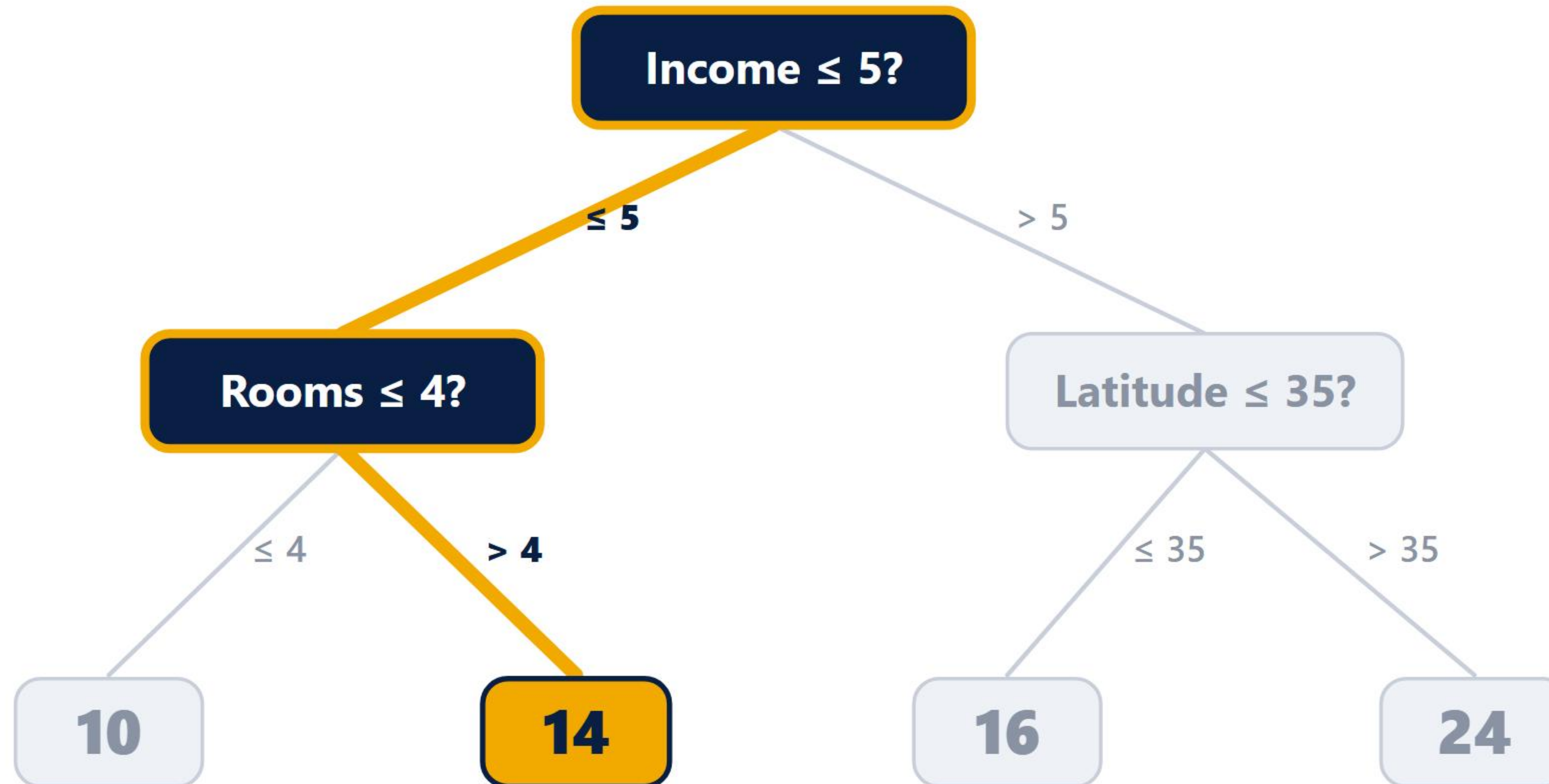
COALITION
S = {L, R}

v(S) = 14

Income = 3 (baseline)

Rooms = 6 (home)

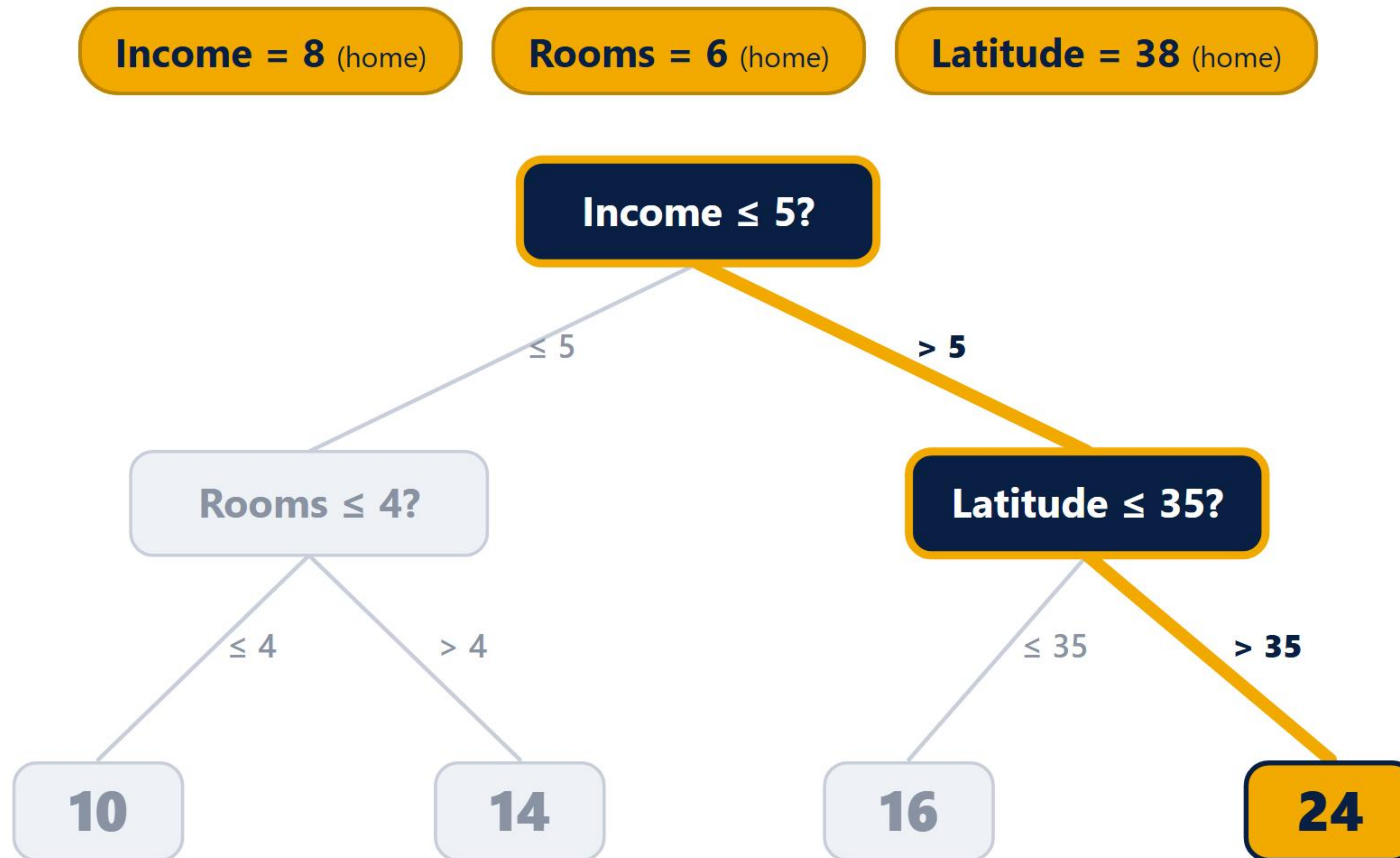
Latitude = 38 (home)



Low income again → only the rooms split matters; latitude is ignored.

COALITION
S = {I, L, R}

v(S) = 24



Everyone known — this leaf IS the model's prediction.

The 8 coalitions → the SHAP values

coalition S	$v(S)$
$\{\}$ (baseline)	10
$\{I\}$	16
$\{R\}$	14
$\{L\}$	10
$\{I, R\}$	16
$\{I, L\}$	24
$\{L, R\}$	14
$\{I, L, R\}$	24

base value $v(\emptyset) = \mathbf{10}$

φ income = **8**

φ rooms = **2**

φ latitude = **4**

$10 + (8 + 2 + 4) = 24 = \text{the prediction } \checkmark$

Each φ is the feature's average boost across all these coalitions.

Why SHAP? The 4 properties that make it fair

- 1 • Efficiency** — the parts add up to the whole: base + all φ = the prediction. ($10 + 8 + 2 + 4 = 24$ ✓)
- 2 • Symmetry** — two features that contribute the same in every coalition get the same φ . Equal work, equal credit.
- 3 • Dummy (null player)** — a feature that never changes any coalition's score gets $\varphi = 0$. No effect, no credit.
- 4 • Additivity** — explanations add across models, so φ for a forest = average of the per-tree φ . This is how TreeExplainer scales up.

SHAP is the ONLY credit split that satisfies all four at once — that uniqueness is why we use it.